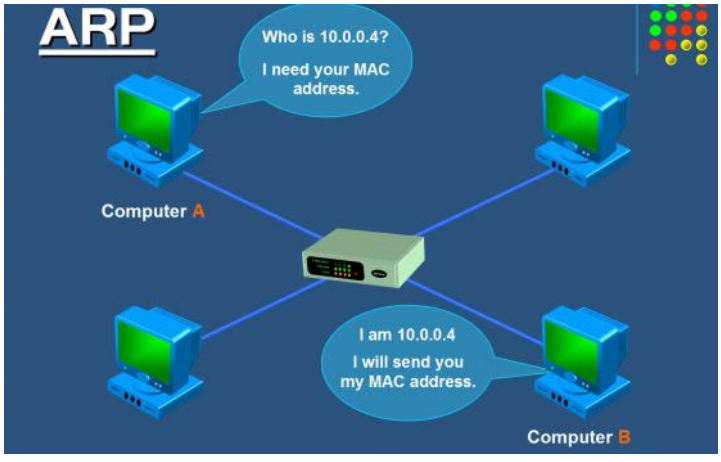
ARP

Saturday, February 28, 2026 8:07 PM

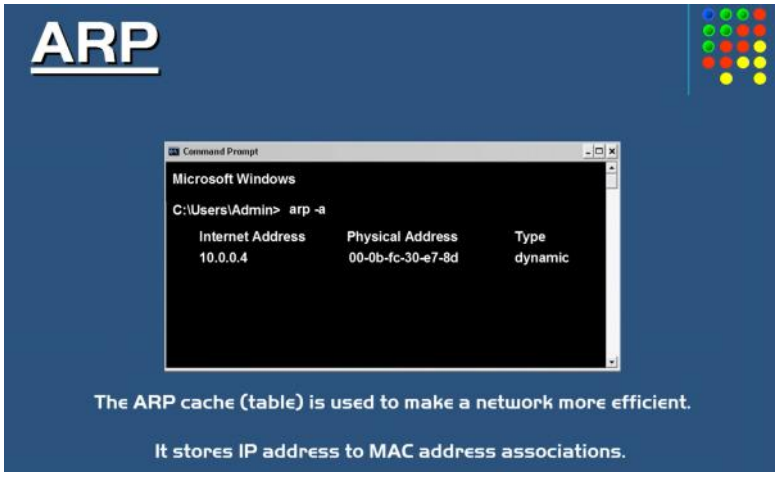
Address Resolution Protocol (ARP) is a fundamental network protocol used to map IP addresses to MAC addresses within a local area network (LAN). This mapping is crucial because while an IP address identifies a device's location on a network, the MAC address uniquely identifies the physical device's network interface card (NIC).

MAC Address: A globally unique physical address assigned to every network interface card.
IP Address: Used to locate a device on a network.
ARP's Purpose: Resolves IP addresses to their corresponding MAC addresses to enable communication between devices on the same LAN.

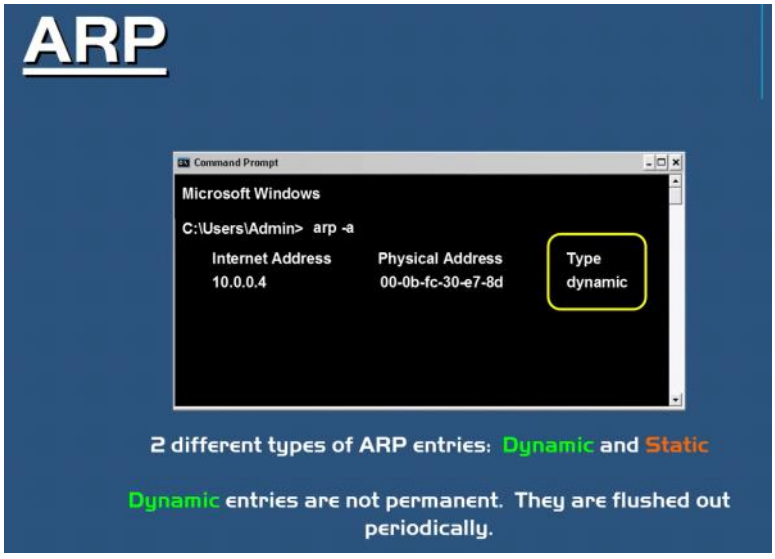


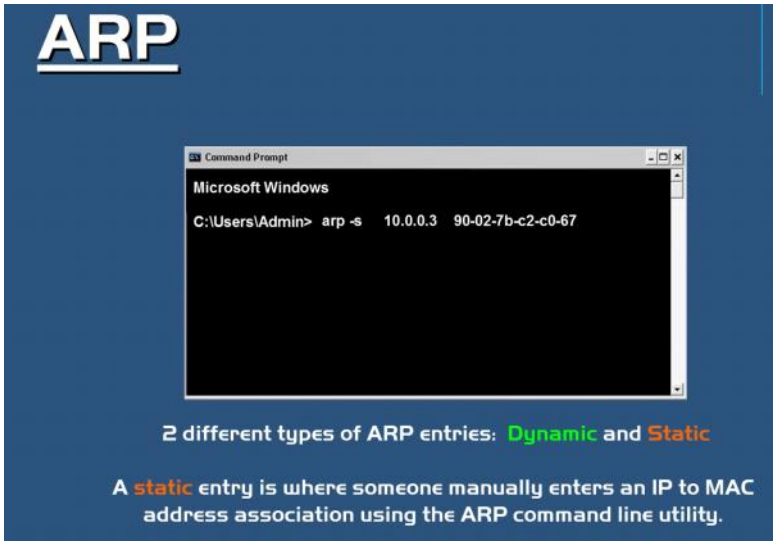
How ARP Works: Step-by-Step Process

Step	Description
1	Computer A wants to communicate with Computer B knowing only Computer B's IP address.
2	Computer A checks its ARP cache (internal list) to find if it already has Computer B's MAC address.
3	If no entry exists, Computer A broadcasts an ARP request to all devices on the network.
4	The device with the matching IP address (Computer B) responds with its MAC address.
5	Computer A receives the MAC address and stores the IP-MAC association in its ARP cache.
6	Communication between Computer A and Computer B proceeds using the resolved MAC address.



ARP Cache
A temporary storage that holds IP-to-MAC address mappings.
Enhances network efficiency by reducing broadcast traffic.
Entries in the ARP cache allow devices to quickly retrieve MAC addresses without broadcasting requests each time.





Types of ARP Entries

Entry Type	Description	Characteristics
Dynamic	Automatically created when a device broadcasts an ARP request and receives a reply.	Temporary; entries are flushed periodically to avoid cache overflow.
Static	Manually created using the arp -s command with specified IP and MAC addresses.	Permanent until manually removed; reduces ARP broadcast traffic.

Use Cases for Static ARP Entries -
Ideal for devices that communicate frequently, to avoid repeated ARP broadcasts.
Network administrators use static entries to optimize network performance and reduce unnecessary traffic.
Example command to create a static ARP entry:
arp -s <IP address> <MAC address>

ARP is essential for local network communication by linking IP addresses to physical MAC addresses.
The ARP cache improves network efficiency by storing resolved addresses, minimizing broadcast requests.
Dynamic ARP entries provide flexibility but require periodic clearing to maintain cache health.
Static ARP entries are strategic tools for network administrators to optimize communication between known devices.
The ARP process operates transparently to end-users but is critical for seamless network operations.